

Project DNA

Build a full-stack Mutual Fund Analytics Platform using publicly available Indian mutual fund data from AMFI India and mfapi.in. Design and implement a complete data pipeline: ingest raw NAV, AUM, and SIP data → clean & load into relational DB → exploratory & performance analytics → interactive Power BI / Tableau dashboard.

10 Real AMCs

SBI MF, HDFC MF, ICICI Prudential, Nippon India, Kotak, Axis, Aditya Birla Sun Life, UTI, Mirae Asset, DSP MF — anchored to AMFI figures (SBI AUM ₹12.5L Cr, industry SIP ₹31,002 Cr Dec 2025).

Data Scale

40 real AMFI scheme codes · 46,000+ daily NAV records · 32,000+ investor transactions · Benchmark indices (Nifty 50, Nifty 100, BSE SmallCap) & portfolio holdings.

Tech Stack

Python 3.10+ · Pandas · NumPy · Matplotlib · Seaborn · Plotly · SQLite · SQLAlchemy · SciPy · Jupyter Lab · Power BI Desktop · Git + GitHub · mfapi.in REST API

Domain

Mutual Fund / Fintech — end-to-end analytics engineering + dashboarding (no API keys required, all public sources).

Core Objectives & Deliverables

#	Objective	Output Artifact
O1	Build Python ETL pipeline from raw AMFI / mfapi.in data	Automated .py script
O2	Design a normalised SQL star schema (5+ tables)	schema.sql
O3	Perform comprehensive EDA on NAV, AUM, SIP data	EDA notebook, 15+ charts
O4	Compute Sharpe, Sortino, Alpha, Beta, VaR, Max Drawdown	Metrics notebook + CSVs
O5	Build a 4-page interactive Power BI / Tableau dashboard	dashboard.pbix / .twbx file
O6	Analyse investor demographics and transaction patterns	Demographic insights

07 Benchmark fund returns vs Nifty 50 / Nifty 100 Benchmark comparison chart

08 Document and present findings PDF report + 12-slide deck

 Provided Datasets (10 real-world files)

File	Rows	Description
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01_fund_master.csv	40	40 real AMFI schemes with codes, fund house, expense ratio, benchmark, fund manager, risk grade
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02_nav_history.csv	46,000	Daily NAV Jan 2022–May 2026, anchored to real mfapi.in values (e.g. HDFC Top 100 Direct code 125497)
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03_aum_by_fund_house.csv	90	Quarterly AUM (₹ crore) for top 10 AMCs 2022–2025 (SBI ₹11.14L Cr Dec 2024)
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04_monthly_sip_inflows.csv	48	Real AMFI Monthly Note — SIP inflows, active accounts Jan 2022–Dec 2025 incl ₹31,002 Cr milestone
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05_category_inflows.csv	144	Net inflows by category (Large Cap, Mid Cap, Small Cap, ELSS, Liquid, Gilt etc.) for FY 2024–25
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06_industry_folio_count.csv	21	Total MF folios (crore) — growth from 13.26 Cr (Jan 2022) to 26.12 Cr (Dec 2025)
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07_scheme_performance.csv	40	1yr/3yr/5yr CAGR, Sharpe, Sortino, Alpha, Beta, Max Drawdown, Std Dev for all 40 schemes
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08_investor_transactions.csv	32,000+	SIP / Lumpsum / Redemption for 5,000 investors across 12 Indian states with age, income, city tier
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09_portfolio_holdings.csv	320	Top stock holdings per equity fund — stock, sector, weight %, market value
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10_benchmark_indices.csv	8,050	Daily close for Nifty 50, Nifty 100, Nifty Midcap 150, BSE SmallCap, CRISIL Liquid & Gilt
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 46k+ NAV records

 AUM data: ₹12.5L Cr SBI anchor

 SIP inflow peak: ₹31,002 Cr (Dec '25)

 5k investors, 12 states

 Data Pipeline & Architecture (Star Schema)

ETL Pipeline (Python)

Extract: mfapi.in REST API + local CSVs (AMFI official)

Transform: pandas cleaning, type casting, handling missing NAVs

Load: SQLite / PostgreSQL with SQLAlchemy ORM

Automated scheduling script: `etl\_pipeline.py`

Normalised Star Schema (5+ tables)

dim_fund (scheme_code, name, AMC, category, risk_grade)

fact_nav (nav_id, scheme_code, nav_date, nav_value)

fact_aum (aum_id, amc_name, quarter_date, aum_crores)

fact_sip (sip_id, month_year, sip_inflow_cr, active_accounts)

fact_transactions (txn_id, investor_id, scheme_code, amount, type)

bridge_benchmark, dim_investor, etc. → fully joinable for BI

 Sample transformation snippet (EDA ready)

```
import pandas as pd
```

```
nav_df = pd.read_csv('02_nav_history.csv')
```

```
perf_df = pd.read_csv('07_scheme_performance.csv')
```

```
merged = nav_df.merge(perf_df, on='scheme_code')
```

```
# compute rolling volatility, Sharpe etc.
```

```
print(merged.groupby('fund_house')['nav_return'].mean())
```

Advanced Analytics — Risk & Performance Metrics

 Risk Metrics: Sharpe Ratio, Sortino Ratio, Value at Risk (VaR 95%), Maximum Drawdown, Beta vs Nifty 50, Alpha (Jensen).

 Performance KPIs: Rolling CAGR (1/3/5Y), Information Ratio, Standard Deviation, Upside Capture.

 Statistical Tests: Normality of returns, correlation matrix of sectoral indices, AUM growth regression.

 EDA Deliverables: 15+ charts: NAV trend heatmaps, SIP account growth, category inflow treemaps, folio count explosion, AUM bar race, scatter of risk-return.

 Output files from metrics notebook: fund_sharpe_ranks.csv
var_drawdown_summary.csv alpha_beta_table.csv

 BI Dashboard (Power BI / Tableau) — 4 Interactive Pages

 Page 1: Market Overview

— Industry AUM, Folio growth, SIP inflows timeline (₹31,002 Cr milestone), category-wise net flows. KPI cards + forecast trend.

 Page 2: Fund Performance & Risk

— Sharpe vs Sortino scatter, Max Drawdown bar chart, fund ranking by Alpha, rolling beta comparison against Nifty 50.

 Page 3: Investor Demographics & Transactions

— Age/income distribution, city tier (Tier1/2/3), SIP vs Lumpsum pie, redemption patterns across 12 states. Choropleth map of India.

 Page 4: Portfolio Holdings & Sector Exposure

— Sunburst of top holdings, sector concentration, weight % by market cap, stock-level exposure across top 5 funds.

 Demo-ready insights: “Mid-cap funds outperformed large-cap by 3.2% alpha (3Y)”, “Highest SIP contribution from Maharashtra & Karnataka”, “Equity oriented AUM growth +41% during 2023-2025”.

 Benchmark & Index Comparison

Compare fund returns against Nifty 50, Nifty 100, Nifty Midcap 150, BSE SmallCap using daily benchmark indices (10_benchmark_indices.csv). Compute tracking error, relative performance line charts, and rolling 6-month correlation. Visual output: Benchmark comparison chart (O7) with interactive tooltips for each scheme.

```
# Example benchmark alignment

import plotly.express as px

# merged_bench = fund_returns.merge(benchmark_df, on='date')

# fig = px.line(merged_bench, title="HDFC Top 100 vs Nifty 50 Rolling Returns")
```

Investor Demographics & Transaction Patterns (O6)

Key Dimensions:

Age bands: 25-34 (highest SIP penetration), 45-60 (high lumpsum)

Income buckets: ₹5-12L, ₹12-25L, ₹25L+ -> correlation with risk appetite

City tier: Tier 1 accounts 56% of total transactions, but Tier 2 growing 2.5x faster

Redemption behaviour: sharp spikes in March & November (tax harvesting?)

 Transaction Mix: from 08_investor_transactions.csv : SIP monthly frequency dominates (68%), systematic withdrawal plans (SWP) emerging in high-net-worth segment. Churn rate by fund house & category.

 States: MH, KA, TN, DL, GJ, UP, WB, etc.

 31% investors under 30 years

 Tier-2 cities: +19% YoY growth

 Milestone & Execution Roadmap (7 working days)

Day Focus

Day 1-2 ETL development: connect to mfapi.in, parse CSVs, load into SQLite (40+ schemes, 46k NAV rows). Validate data quality.

Day 3 Star schema design + indexing. Write schema.sql and full database setup script. Perform initial EDA with pandas/profiling.

Day 4 Risk & performance analytics notebook: Sharpe, Sortino, VaR, MaxDD, alpha/beta, generate metric tables.

Day 5 Build interactive Power BI/Tableau dashboard (4 pages + slicers). Connect live to aggregated views.

Day 6 Demographic deep dive, benchmark comparison, fine-tuning visuals, and prepare slide deck outline.

Day 7 Documentation, final PDF report, 12-slide presentation, dashboard publishing, and GitHub repo commit.

✅ All data sources are free & publicly accessible — AMFI India, mfapi.in, NSE/BSE public indices. No API keys required, perfect for capstone showcase.

References & Resources

 mfapi.in REST API docs

 AMFI monthly SIP report (Dec 2025)

 GitHub repo structure: /etl, /notebooks, /dashboard, /docs

 Power BI theme: fintech blue+emerald

⚡ Capstone final submission: include full Python pipeline, interactive dashboard, and a recorded walkthrough highlighting SIP surge (₹31,002 Cr milestone) and performance of top 5 equity funds vs Nifty 100.

🧠 Bluestock Fintech Capstone — Mutual Fund Analytics Platform · Data sourced from AMFI India, mfapi.in, and NSE/BSE public records · For internal evaluation & portfolio showcase.